

3_Lists

September 15, 2026

```
[1]: mylist=[1,2,3,4,5]
      print(mylist)
```

```
[1, 2, 3, 4, 5]
```

```
[2]: mylist2=[1,'a','3.5','Hanu']
      print(mylist2)
```

```
[1, 'a', '3.5', 'Hanu']
```

```
[3]: print(mylist2[0])
```

```
1
```

```
[4]: print(mylist2)
```

```
[1, 'a', '3.5', 'Hanu']
```

```
[7]: mylist2[0]=5 #List is mutable, we can change the value of an element in a list
```

```
[6]: print(mylist2)
```

```
[5, 'a', '3.5', 'Hanu']
```

```
[8]: #List slicing
```

```
print(mylist2[0:3]) #Prints elements from index 0 to 2
```

```
[5, 'a', '3.5']
```

```
[9]: #List reversal
      mylist2.reverse()
      print(mylist2)
```

```
['Hanu', '3.5', 'a', 5]
```

```
[10]: print(mylist2)
       rev_list=mylist2[::-1] #Another way to reverse a list
       print(rev_list)
```

```
['Hanu', '3.5', 'a', 5]
```

```
[5, 'a', '3.5', 'Hanu']
```

```
[13]: # Modifying a list
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```
print(mylist)
mylist.append(6)
mylist.remove(3)
print(mylist)
```

```
[1, 2, 3, 4, 5]
```

```
[1, 2, 4, 5, 6]
```

```
[14]: #concatenation of lists
```

```
mylist3 = mylist + mylist2
print(mylist3)
```

```
[1, 2, 4, 5, 6, 'Hanu', '3.5', 'a', 5]
```

```
[16]: #list repetition
```

```
mylist4 = [1] * 10
print(mylist4)
```

```
[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
```

```
[17]: print(len(mylist4)) #length of the list
```

```
10
```

```
[ ]: #List within a list or nested list
```

```
mylist5 = [[1, 2], [3, 4,5], [6,7,8,9], ['Hanu',37,'200k']]
print(mylist5)
```

```
[[1, 2], [3, 4, 5], [6, 7, 8, 9], ['Hanu', 37, '200k']]
```

```
[24]: #fetching an element from a list within a list
```

```
print(mylist5[1][2]) #fetching 3rd element from 2nd list
print(mylist5[3][0]) #fetching 1st element from 4th list
```

```
5
```

```
Hanu
```

```
[30]: # List element manipulation
```

```
print(mylist2)
mylist2.append(7) #Appends 7 to the end of the list
mylist2.insert(2,10) #Inserts 10 at index 2
mylist2.pop(1) #Removes the element at index 1
mylist2.remove('a') #Removes the first occurrence of '3.5'
print(mylist2)
```

```
['Hanu', 10, 10, 'a', 5, 7, 7, 7, 7]
```

```
['Hanu', 10, 10, 5, 7, 7, 7, 7, 7]
```

```
[31]: # List comprehension
```

```
squares = [x**2 for x in range(10)]
```

```
print(squares)
```

```
[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```